24 Multiple Reactions with Heat Effects

Recall the energy balances that we've been looking at so far, where we have:

- Heat Generated term (Q_g or G(T)) -- due to the reaction
- Heat Removed term (Q_r or R(T)) -- due to the heat exchanger and also sensible heat changes as reactor contents "absorb" heat given off by the reaction

The Heat Generated terms look like: $(r_A)(\Delta H^{rxn})$ ← must be in units of energy / mol A

For the case of multiple reactions, the Heat Generated term looks like: $\sum_i(-r_{ij})(-\Delta H_{ij}^{rxn})$, where *i* are reactions and *j* are species.

In this sum, there should be **one $(-r_j)(-\Delta H_j^{rxn})$ term per reaction**. The purpose of the *j* subscript is to ensure consistent stoichiometry/units ... that each rate of reaction and heat of reaction pair are taken with respect to the same species. What matters here is that *j* is consistent for each pair; it is OK if *j* varies from pair to pair.

Beyond that, the solution procedure is analogous to the one you've already used to solve reactor design problems when there are multiple reactions: You have to use *N*'s or *F*'s instead of *X*'s, and because of this you need to use mathematical software.

Let's do an example:

The gas phase elementary reactions 2A ⇋ B and 2B + A --> C are carried out in a non-isothermal PFR with no pressure drop. Write the symbolic expression for the heat generated term.

Info:

k_{1A} = 50 M^{-1} min^{-1} at 350 K with E = 8 kJ/mol
k_{2C} = 4000 M^{-2} min^{-1} at 310 K with E = 4 kJ/mol
$K_{C,A}$ = 10 M^{-1} at 315 K
ΔH_{1A}^{rxn} = -20 kJ/mol A
ΔH_{2B}^{rxn} = +30 kJ/mol B

rate of cons if reactant
& rate of gen if product

$$\left(\frac{\text{moles A}}{L \cdot min}\right)\left(\frac{kJ}{mol\,A}\right) = \frac{kJ}{L \cdot min}$$

$$\frac{dT}{dV} = \frac{Q_g + Q_r}{\sum F_i C_{pi}}$$

$$Q_g = \sum_i (-r_{ij})(-\Delta H_{ij}^{rxn}) = (-r_{1A})(-\Delta H_{1A}^{rxn})$$

$$+ (r_{2C})\underbrace{(-\Delta H_{2B}^{rxn})\left(\frac{2\text{ mol B}}{1\text{ mol C}}\right)}_{2(-\Delta H_{2B}^{rxn}) = -\Delta H_{2C}^{rxn}}$$

ΔH_{1A}^{rxn}: $-20\frac{kJ}{mol\,A}$; $-\Delta H_{1A}^{rxn}$: $+20\frac{kJ}{mol\,A}$

$$\frac{r_{2C}}{1} = \frac{r_{2B}}{-2}$$

$$r_{2C} = -\frac{1}{2} r_{2B} \longrightarrow \underbrace{\left(-\frac{1}{2} r_{2B}\right)}_{\frac{1}{2} k_{2C} C_B^2 C_A}\left(-\underbrace{\Delta H_{2B}^{rxn}}_{+30\frac{kJ}{mol\,B}}\right)$$

$-30\frac{kJ}{mol\,B}$

use Arrhenius expression to determine as a function of T.

use stoichiometry expressions for concentration

$$C_j = C_{T0}\left(\frac{F_T}{F_{T0}}\right)\left(\frac{T_0}{T}\right)$$

$$F_T = F_A + F_B + F_C$$

using matrices

(1) make sure units on ΔH^{rxn} are consistent w/ units on rate.

rxn 1: $2A \rightleftarrows B \quad -r_{1A} = k_{1A}\left(C_A^2 - \frac{C_B}{K_{CA}}\right) \quad \Delta H^{rxn}_{1A} = -20 \frac{kJ}{mol\ A}$ ✓

rxn 2: $2B + A \rightarrow C \quad r_C = k_{2C} C_B^2 C_A \quad \Delta H^{rxn}_{2B} = 30 \frac{kJ}{mol\ B}$ ✗

$$\Delta H^{rxn}_{2C} = \left(30 \frac{kJ}{mol\ B}\right)\left(\frac{2\ mol\ B}{mol\ C}\right) = 60 \frac{kJ}{mole}$$

(2) break reversible rxns up:

new rxn 1: $2A \rightarrow B$ $\quad -r_{1A} = k_{1A} C_A^2$

new rxn 2: $B \rightarrow 2A$ $\quad r_{2A} = \frac{k_{1A}}{K_{CA}} C_B$

new rxn 3: $2B + A \rightarrow C$ $\quad r_{3C} = k_{3C} C_B^2 C_A$

(3) create array of ΔH rxns

$$H = \begin{bmatrix} -20 \\ 20 \\ 60 \end{bmatrix}$$

(4) Solve for Q:

$$Q_g = r(-H^T)$$

$$r = \begin{bmatrix} k_{1A} C_A^{2} \\ \frac{k_{1A}}{K_{CA}} C_B \\ k_{3C} C_B^2 C_A \end{bmatrix}$$

$$r(-H^T) = \begin{bmatrix} k_{1A} C_A^2 \\ \frac{k_{1A}}{K_{CA}} C_B \\ k_{3C} C_B^2 C_A \end{bmatrix} \begin{bmatrix} 20 & -20 & -60 \end{bmatrix}$$

$$Q_g = \left(20 \frac{kJ}{mol\,A}\right)\left(k_{1A} C_A^2\right) - \left(20 \frac{kJ}{mol\,A}\right)\left(\frac{k_{1A}}{K_{CA}} C_B\right)$$

$$- \left(60 \frac{kJ}{mol\,C}\right) k_{3C} C_B^2 C_A$$